

SPECIFICATION — NODE 3 — PAGE 1

EXTERNAL SESSION-BASED GEOMETRY SCANNER (C-SHAPE OR CLOSED RING) FOR BASELINE LENGTH/DIAMETER CHARACTERIZATION WITH VALIDATION AND ANTI-SPOOF (NON-LIMITING)

A. Technical Field

This section relates to external, session-based geometry scanning devices configured to generate baseline geometry references using time-series sensing during a user-performed scan. The disclosed embodiments further relate to validity gating, artifact rejection, and anti-spoof mechanisms for improving robustness of baseline geometry characterization and for generating privacy-preserving outputs suitable for standalone use and optional correlation with other nodes.

B. Summary (Non-Limiting)

In various embodiments, an external session-based geometry scanner is configured as a C-shaped device or a substantially closed ring and is used in a discrete scan workflow to generate a baseline geometry reference. The geometry scanner acquires measurements indicative of length using one or more optical flow sensors and indicative of diameter using one or more distance sensors including time-of-flight (ToF), LiDAR, and/or infrared (IR) distance sensing. The device further includes artifact rejection and validation sensing including a 6-axis IMU and may include anti-spoof and physiologic confirmation sensing including PPG and/or pressure/contact sensing (e.g., force sensing resistor). In certain embodiments, the device further includes temperature sensing to provide thermal context, contact plausibility checks, and/or compensation for sensor drift. A computing system receives sensor signals, computes quality indices, selects valid segments, and outputs privacy-preserving baseline references comprising indices, zones, ranges, and confidence indicators.

C. Exemplary Physical Embodiments (Non-Limiting)

In various embodiments, the geometry scanner comprises a housing forming a C-shaped body with a gap or a substantially closed ring. The device includes one or more sensing regions arranged to observe relative motion and distance during a scan path. Non-limiting embodiments include: 1. a C-shaped scanning ring configured to be placed at a starting position and moved along a scan direction; 2. a substantially closed scanning ring configured to constrain motion and improve repeatability; and/or 3. a multi-sensor ring with distributed sensing modules positioned around the inner circumference. In certain embodiments, the device includes alignment cues and a user guidance workflow (e.g., on-device feedback and/or app guidance) to support repeatable scanning.

D. Sensor Suite (Non-Limiting)

In various embodiments, the geometry scanner includes one or more sensors including, without limitation: 1. optical flow sensing (one or more sensors) configured to estimate relative motion and/or incremental distance along a scan path; 2. distance sensing for diameter estimation, including one or more of ToF sensors, LiDAR, and/or IR distance sensors; 3. inertial sensing, including a 6-axis IMU configured for orientation tracking, motion gating, and artifact rejection; 4. contact/pressure sensing, including FSR or other pressure sensors configured to confirm coupling and stabilize validity; 5. optical physiologic confirmation, including PPG in certain embodiments configured to improve anti-spoof confidence; and/or 6. temperature sensing, including one or more thermal sensors configured to provide thermal context,

coupling plausibility, and/or compensation for temperature-dependent drift. Any sensor modality may be substituted with an equivalent modality providing substantially similar function. Sensors may be described as sensing regions rather than fixed placements to support design variation.

E. Collection Modes and Scan Workflow (Non-Limiting)

In various embodiments, the geometry scanner is used in a discrete scan session comprising: 1. placement and contact confirmation; 2. acquisition of sensor streams during a scan motion; 3. validation scoring and segment selection; and 4. generation of a baseline geometry reference with confidence indicators. In certain embodiments, multiple scans are performed and aggregated to improve repeatability and confidence.

F. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 3 generates privacy-preserving baseline geometry references and repeatability/confidence scores for a single user without requiring other nodes. In optional multi-node embodiments, baseline references may be used to normalize longitudinal metrics and/or support cross-device correlation outputs under consent gating and quality thresholds.